Final Project Proposal: AI Bootcamp

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- **Submission Date:** October 3, 2026
- **Selected Track:** Natural Language Processing (NLP) / Large Language Models (LLMs)
- **Project Title: AI-Powered Semantic Product Discovery, Personalized Recommendation & Conversational Shopping Concierge**

1. Project Idea and Objective

Traditional e-commerce product discovery often relies heavily on keyword matching, which can perform poorly when users express their needs using natural, descriptive language and implicit constraints, such as *"breathable running jacket for rainy weather under $80."*

The objective of this project is to build an end-to-end AI-powered product discovery and recommendation system that understands natural-language shopping queries, retrieves semantically relevant products, applies deterministic metadata constraints, personalizes product ranking using user interaction history, and generates grounded conversational recommendations using an LLM.

The system will combine semantic information retrieval, lightweight personalization, deterministic filtering, and retrieval-augmented generation (RAG) into a single prototype for an individual e-commerce catalog.

2. Problem Statement

Modern e-commerce product discovery can face several limitations:

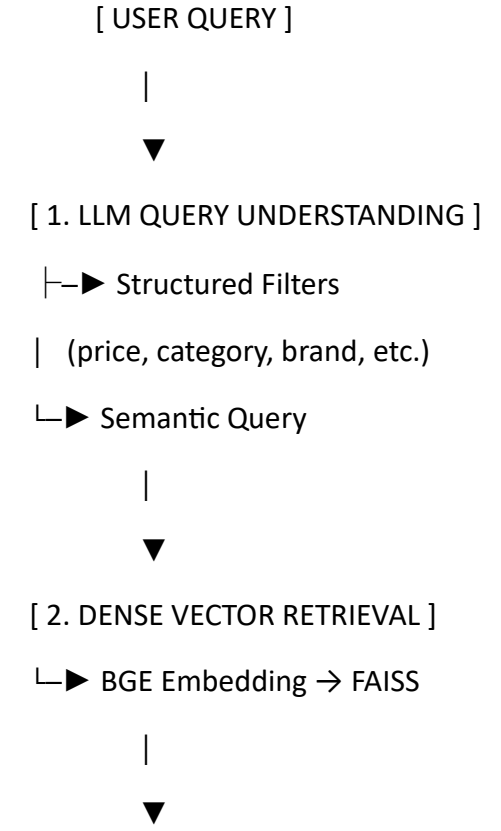
- **Vocabulary Mismatch:** Keyword-based search depends heavily on overlapping terms and may miss semantically relevant products described using different language. For example, a query for "breathable rain jacket" may fail to retrieve a product described as a "lightweight waterproof shell with moisture-wicking fabric."
- **Constraint Handling:** Semantic similarity alone does not reliably enforce strict user constraints such as maximum price, category, or brand. A semantically similar product may still violate an explicit requirement such as *"under $80."*
- **Limited Personalization and Explanation:** Standard search and recommendation interfaces may not sufficiently incorporate a user's previous interactions or explain why particular products are relevant to the user's current request.

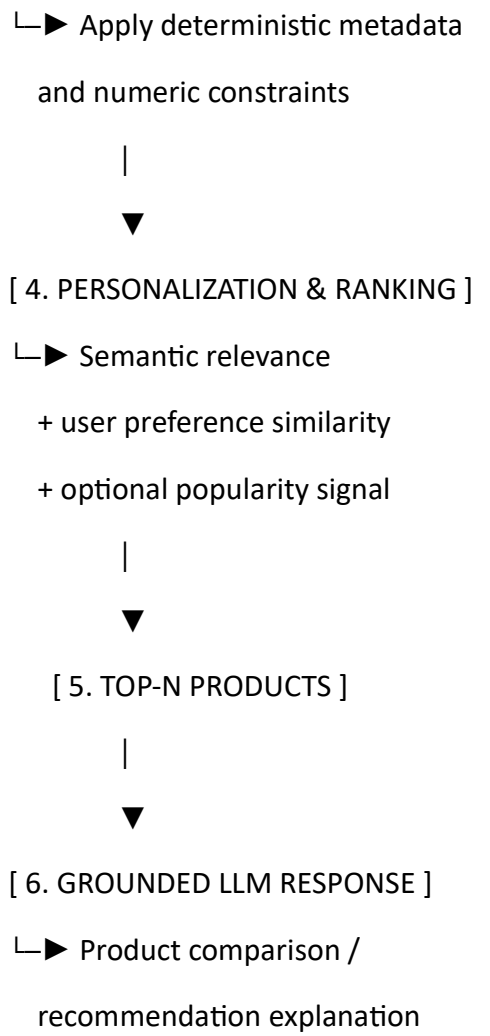
The proposed system addresses these challenges by combining semantic retrieval, structured filtering, personalization, and grounded conversational generation.

3. Why the Problem Is Important and Relevant

- **Product Discovery:** More relevant search and recommendation results can reduce friction in product discovery and help shoppers identify suitable products more efficiently.
- **Relevance to the NLP/LLM Track:** The project applies Natural Language Processing and Information Retrieval concepts including natural-language query understanding, dense semantic embeddings, vector similarity search, structured information extraction, and retrieval-augmented generation.
- **End-to-End Scope:** The project covers the workflow from product catalog ingestion and semantic indexing to retrieval, ranking, evaluation, and an optional interactive demonstration, satisfying the end-to-end nature of the final project.

4. High-Level Approach and Architecture





1. Query Understanding

An instruction-tuned LLM parses the user's natural-language request into structured filters such as max_price, min_price, category, and brand, together with a cleaned semantic query.

The LLM is responsible for **extracting** the constraints; deterministic Python logic is responsible for **enforcing** them.

2. Dense Semantic Retrieval

The semantic query is converted into a dense vector using the BAAI/bge-small-en-v1.5 sentence-embedding model.

The resulting vector is searched against an in-memory FAISS index to retrieve a broad candidate pool, such as the Top-100 or Top-200 semantically similar products.

3. Deterministic Post-Filtering

FAISS candidate IDs are mapped to the corresponding rows in the Pandas product catalog.

Explicit constraints such as maximum price and category are then applied using deterministic Python/Pandas filtering to remove products that violate the user's requirements.

4. Personalization and Ranking

The remaining products are reranked using a lightweight hybrid scoring function that combines semantic relevance with similarity to a user preference representation derived from previous interactions.

For example:

$$Score(p) = w_1 S_{semantic}(p) + w_2 S_{personal}(p) + w_3 S_{popularity}(p)$$

where:

- $S_{semantic}$ represents similarity between the query and product embedding.
- $S_{personal}$ represents similarity between the product and the user's preference representation.
- $S_{popularity}$ is an optional fallback signal based on product interaction frequency.

5. Top-N Recommendation

The highest-ranked products are selected for the final recommendation set, typically 3–5 products.

6. Grounded Conversational Synthesis

The selected products and their verified catalog attributes are provided to an LLM as retrieved context.

The LLM generates a concise shopping response or product comparison grounded only in the retrieved product information.

5. Evaluation and Quantitative Benchmark Strategy

The system will be evaluated using a curated benchmark of approximately 25 diverse natural-language shopping queries.

Baselines for Comparison

1. **TF-IDF Keyword Search:** Traditional lexical similarity baseline using TF-IDF and cosine similarity.

2. **Dense Semantic Search:** BGE embeddings with FAISS retrieval, without metadata filtering or personalization.
3. **Proposed Hybrid Pipeline:** LLM query understanding + dense semantic retrieval + deterministic metadata filtering + lightweight personalization/ranking.

Evaluation Metrics

- **Precision@K:** Proportion of retrieved products that are relevant to the query.
- **Recall@K:** Proportion of relevant products successfully retrieved.
- **NDCG@K:** Ranking quality of the retrieved relevant products.
- **Constraint Adherence Rate:** Percentage of final recommendations satisfying explicitly requested constraints such as price and category.
- **End-to-End Latency:** Execution time for retrieval, filtering, ranking, and conversational response generation.

Qualitative Analysis

Error analysis will examine:

- semantic retrieval failures
- incorrect or incomplete query-filter extraction
- insufficient catalog information
- products incorrectly removed by hard constraints
- personalization errors
- LLM grounding and response quality

6. Technical Stack and Tools

- **Language:** Python 3.11+
- **Data Processing:** Pandas, NumPy
- **Machine Learning / Evaluation:** Scikit-learn
- **Semantic Embeddings:** Sentence Transformers with BAAI/bge-small-en-v1.5
- **Vector Search:** FAISS-CPU
- **LLM:** OpenAI API or Google Gemini API

- **Environment:** Google Colab / Jupyter Notebook
- **Visualization:** Matplotlib, Seaborn
- **Optional Demonstration Interface:** Streamlit

7. Scope and Feasibility

The project will focus on a prototype for a single e-commerce catalog and will prioritize completion of the core retrieval, filtering, personalization, and grounded conversational pipeline.

The following capabilities will be added later after version 1 of this project completes:

- multi-tenant architecture
- Shopify/WooCommerce integrations
- real-time event streaming
- production-scale deployment
- LLM fine-tuning
- autonomous purchasing
- large-scale distributed infrastructure

A Streamlit/Universal Store connector demonstration and additional personalization capabilities will be implemented only if sufficient time remains after the core pipeline and evaluation are complete.

8. Expected Outcome

The final project will produce an end-to-end prototype capable of interpreting natural-language shopping queries, retrieving semantically relevant products, enforcing explicit product constraints, incorporating basic user preferences, and generating grounded conversational recommendations.

The evaluation will compare traditional lexical retrieval, dense semantic retrieval, and the proposed hybrid approach to determine how each stage affects product relevance, ranking quality, and constraint adherence.